

Automata Theory MCQs (Set-2) – 50 Questions

1. Which of the following is a Type-2 grammar in the Chomsky hierarchy?
A) Regular Grammar
B) Context-Free Grammar
C) Context-Sensitive Grammar
D) Unrestricted Grammar
2. Which automaton accepts Context-Free Languages?
A) DFA
B) NFA
C) PDA
D) TM
3. A grammar is called ambiguous if:
A) It has no derivation
B) It generates infinite strings
C) A string has more than one parse tree
D) It is left recursive
4. Which of the following is a regular language?
A) $\{a^n b^n \mid n \geq 0\}$
B) $\{ww \mid w \in \{a,b\}^*\}$
C) $(ab)^*$
D) $\{a^n b^n c^n \mid n \geq 0\}$
5. Which theorem is used to convert FA into Regular Expression?
A) Pumping Lemma
B) Rice's Theorem
C) Arden's Theorem
D) Myhill-Nerode Theorem
6. The language accepted by a PDA using final states is:
A) Always Regular
B) Context-Free
C) Context-Sensitive
D) Recursive
7. Which of the following is NOT a closure property of regular languages?
A) Union
B) Intersection
C) Complement
D) Equal Number of a's and b's
8. In DFA, the transition function maps:
A) $Q \times \Sigma \rightarrow Q$
B) $Q \times \Sigma \rightarrow 2^Q$
C) $Q \rightarrow \Sigma$
D) $\Sigma \times Q \rightarrow Q$

9. Which automaton can move its head both left and right?
- A) DFA
 - B) NFA
 - C) PDA
 - D) TM
10. The language generated by $S \rightarrow aS \mid bS \mid \epsilon$ is:
- A) $a^n b^n$
 - B) All strings over $\{a, b\}$
 - C) Only a's
 - D) Only b's
11. Which of the following is NOT a component of PDA?
- A) Stack
 - B) Input Alphabet
 - C) Tape Alphabet
 - D) Transition Function
12. A Moore machine produces output based on:
- A) Current Input
 - B) Current State
 - C) Stack Content
 - D) Transition
13. Which machine usually requires fewer states?
- A) Moore
 - B) Mealy
 - C) DFA
 - D) PDA
14. The language $\{a^n b^n c^n \mid n \geq 1\}$ is:
- A) Regular
 - B) CFL
 - C) CSL
 - D) Finite
15. Which of the following is accepted by an NPDA but not by a DPDA?
- A) a^*
 - B) $(ab)^*$
 - C) $\{ww^R\}$
 - D) a^n
16. The empty string is represented by:
- A) $\emptyset$
 - B) λ or ϵ
 - C) Φ
 - D) Σ
17. The length of string "abba" is:
- A) 2
 - B) 3
 - C) 4
 - D) 5

18. Σ^* contains:
- A) Only ϵ
 - B) Infinite strings but not ϵ
 - C) All strings including ϵ
 - D) Only finite strings of length 1
19. Which operation combines two languages sequentially?
- A) Union
 - B) Complement
 - C) Concatenation
 - D) Difference
20. The reverse of "ababa" is:
- A) aabab
 - B) ababa
 - C) baaab
 - D) babaa
21. Which grammar type is accepted by LBA?
- A) Type 0
 - B) Type 1
 - C) Type 2
 - D) Type 3
22. A language accepted by a TM and always halts is:
- A) Recursive
 - B) RE only
 - C) CFL
 - D) Regular
23. Rice's Theorem deals with:
- A) Regular Languages
 - B) Non-trivial properties of RE languages
 - C) PDA acceptance
 - D) DFA minimization
24. Which problem is famous for being undecidable?
- A) DFA Acceptance
 - B) NFA Acceptance
 - C) Halting Problem
 - D) DFA Minimization
25. A Universal Turing Machine simulates:
- A) DFA only
 - B) PDA only
 - C) Any TM
 - D) NFA only
26. Which symbol represents the empty set?
- A) ϵ
 - B) λ
 - C) $\emptyset$
 - D) Σ

27. The grammar $S \rightarrow Sa \mid b$ contains:
- A) Right recursion
 - B) Left recursion
 - C) Ambiguity
 - D) Unit production
28. The purpose of DFA minimization is:
- A) Increase states
 - B) Remove input symbols
 - C) Reduce equivalent states
 - D) Convert DFA to NFA
29. Which language is NOT regular?
- A) $(a+b)^*$
 - B) ab
 - C) $\{a^n b^n\}$
 - D) $(ab)^*$
30. Pumping Lemma is generally used to:
- A) Construct DFA
 - B) Prove non-regularity
 - C) Convert CFG to PDA
 - D) Minimize DFA
31. In a parse tree, leaves are:
- A) Variables
 - B) Start Symbols
 - C) Terminals
 - D) Productions
32. Which of the following is a regular expression?
- A) $a+b$
 - B) $a \rightarrow b$
 - C) $a:=b$
 - D) $a=b$
33. The language accepted by DFA is always:
- A) Recursive
 - B) Regular
 - C) CFL
 - D) CSL
34. Which of the following automata has no memory?
- A) DFA
 - B) PDA
 - C) TM
 - D) LBA
35. The stack in PDA follows:
- A) FIFO
 - B) LIFO
 - C) Random Access
 - D) Circular

36. Which operation is represented by '+' in regular expressions?
- A) Concatenation
 - B) Closure
 - C) Union
 - D) Complement
37. A right-linear grammar generates:
- A) CFL
 - B) CSL
 - C) Regular Languages
 - D) RE Languages
38. Which machine recognizes Context-Sensitive Languages?
- A) PDA
 - B) DFA
 - C) LBA
 - D) NFA
39. A DFA must have:
- A) At least one final state
 - B) Exactly one transition for each input symbol
 - C) ϵ -transitions
 - D) A stack
40. The complement of a regular language is:
- A) Always regular
 - B) Never regular
 - C) Context-Free
 - D) Context-Sensitive
41. Which language class is the most powerful?
- A) Regular
 - B) CFL
 - C) CSL
 - D) RE
42. The language generated by $S \rightarrow ab$ is:
- A) $\{a,b\}$
 - B) $\{ab\}$
 - C) $\{abab\}$
 - D) $\{\epsilon\}$
43. Which of the following is a terminal symbol?
- A) S
 - B) A
 - C) a
 - D) B
44. Which of the following is a non-terminal symbol?
- A) a
 - B) b
 - C) 0
 - D) S

45. In NFA, a state may have:
- A) Exactly one transition
 - B) Multiple transitions for same input
 - C) No states
 - D) No alphabet
46. The Chomsky hierarchy starts from:
- A) Type 0
 - B) Type 1
 - C) Type 2
 - D) Type 3
47. Which machine is more powerful?
- A) DFA
 - B) PDA
 - C) TM
 - D) NFA
48. The language of all strings ending with "01" is:
- A) Finite
 - B) Regular
 - C) Context-Sensitive only
 - D) Not Regular
49. Acceptance by empty stack is associated with:
- A) DFA
 - B) NFA
 - C) PDA
 - D) TM
50. Which statement is correct?
- A) Regular $\supset$ CFL
 - B) CFL $\supset$ Regular
 - C) DFA $\supset$ TM
 - D) PDA $\supset$ TM